

Work instructions

TD10008403 EN 03

Scope of validity	MO-PlasT / MO-ILT / MO-FT / MO-BCT
Area of application	Commissioning the Contiform Block BF and ErgoBloc L

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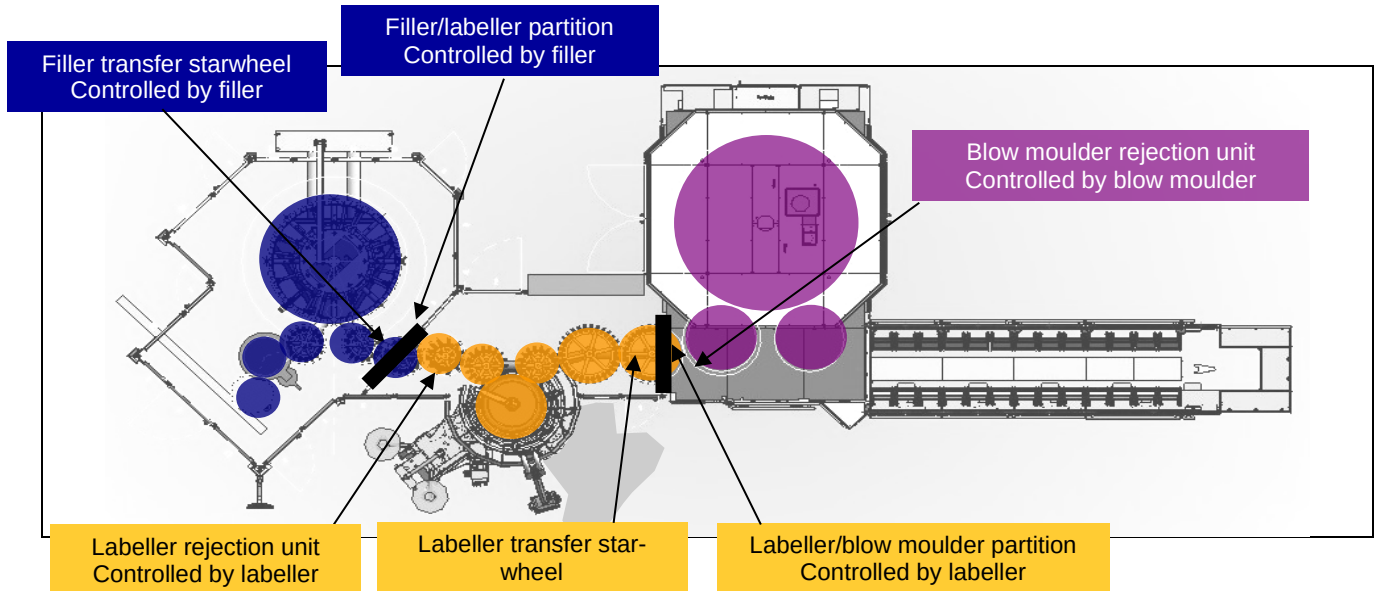
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General information

Overview of the layout and terms



References

- TD10002692 Block FB230 - S7 Functional description
- TD10008505 Request synchronous operation FB222 - S7 Functional description
- TD10008404 Teaching of the shift register positions FB219 - S7 Functional description
- TD10002690 Servo joint control - Service instructions
- TD10003821 Time synchronisation - NTP time server - Service instructions
- ESW0004239 Word document: Signal exchange DP-DP
- TD12000765 Provision of test containers in the labeller

Parameter list single/block:

Krones Inside → Departments → Bottling and packaging equipment → Bottling technology
 → Block and conveyor technology → Block technology → Engineering and order processing block
 → On the right under Download, open the file "[Parameter list single/block](#)" (note the series version)

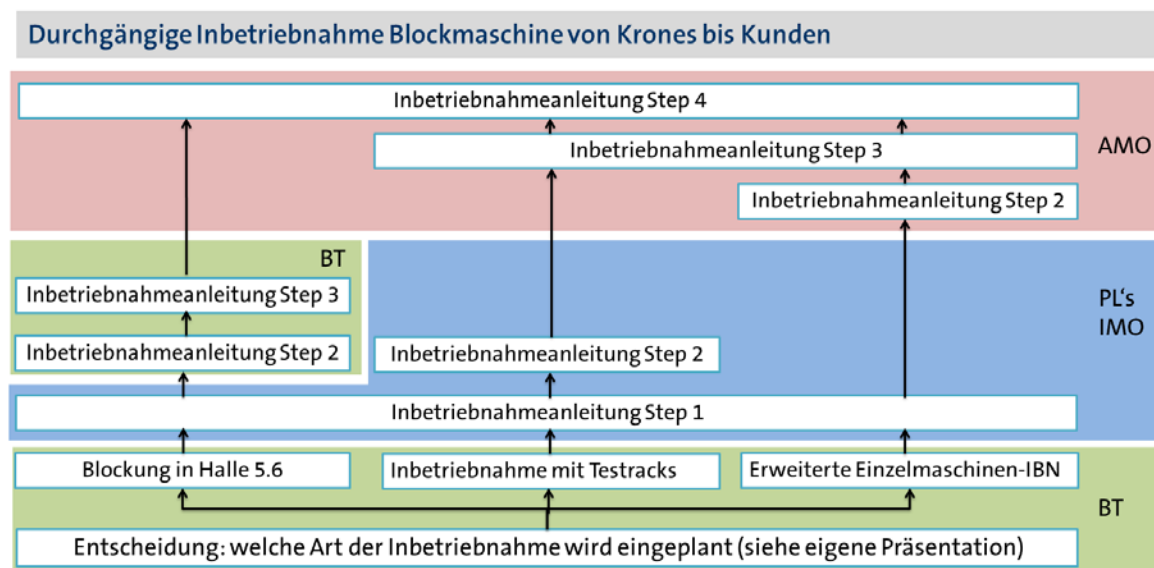
Revision	Date	Comments
01	March 2017	Documentation completely revised

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Responsibilities/procedure

No.	Test	Block-synchronising	Test rack	EEIB
1.1	Safety technology	IMO PL	IMO PL	IMO PL
1.2	Drive technology	IMO PL	IMO PL	IMO PL
1.3	Control technology	IMO PL	IMO PL	IMO PL
1.4	Visualisation system	IMO PL	IMO PL	IMO PL
1.5	Time synchronisation	IMO PL	IMO PL	IMO PL
2.1	Block, general Safety	IMO Block	IMO PL	AMO
2.2	BCM	IMO Block	IMO PL	Not necessary
2.3	Signal exchange	IMO Block	IMO PL	Not necessary
2.4	JOG block	IMO Block	IMO PL	Not necessary
2.5	SJC	IMO Block	IMO PL	Not necessary
2.6	Operating conditions/signal beacon post	IMO Block	IMO PL	Not necessary
2.7	Lock movement	IMO Block	IMO PL	Not necessary
2.8	Labeller guards	IMO Block	IMO PL	Not necessary
3.1	BCM server	IMO Block	AMO	AMO
3.2	Block, general Safety	IMO Block	AMO	Already under 2.1
3.3	Shift register	IMO Block	AMO	AMO
3.4	Partition	IMO Block	AMO	AMO
3.5	Empty situation	IMO Block	AMO	AMO
3.6	Power failure	IMO Block	Not necessary	Not necessary
3.7	Horn/lack of POM (processing and operating materials)	IMO Block	Not necessary	Not necessary
4.1	Block, general Safety	AMO	Already under 3.2	Already under 2.1
4.2	Torques	AMO	AMO	AMO



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1 Step 1: EEIB Commissioning of further individual machines

1.1 Safety technology

Test of the safety technology according to the valid AS-I commissioning manual of the respective PL with measurement of the potentials at the signal exchange terminals or removal of bridges at the inputs

1.2 Drive technology

Observe the SDC commissioning manual

Parameter settings according to the valid default values: see the "Block parameter list"

Following the test phase and before delivery, adjust the temporary settings (e.g. for the transfer starwheel) with respect to the block values

Make the adjustment for the last axis (discharge starwheel blowing wheel axis 4, discharge starwheel labeller axis 8)

Check the SDC version (note the series version)

1.3 Control technology

All:

Check whether the DP-DP coupler(s) for the signal exchange is(are) present in the hardware configuration of the S7 project

Make sure that the FB230 corresponds to the series version of the block machines

Check whether the signal exchange among the block machines is effected according to the signal exchange list of the LDG drive (greying out)

Blow moulder:

Setting of the parameters for "Fine clock pulses until transfer clear": see the "Block parameter list"

Adjust the parameters for "Machine empty": Preform parameters at the infeed P.E. sensor + parameters for the clock pulse for the transfer of the shift register information + 5 clock pulses

Labeller:

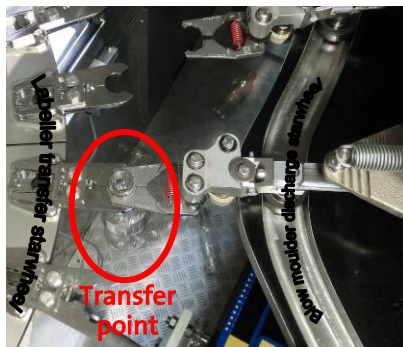
Check whether assembly 36 is present in the blow moulder plan (base cooling present); if so, set configuration bit DB 500 at the labeller

Set the parameters for splicing the label (only block and AMO).

Parameters "maximum no. of containers in the blow moulder" (only block and AMO)

Set the parameters "Transfer starwheel empty" and "Fine clock pulses until transfer clear": see the "Block parameter list"

Set the parameters for "Machine empty": Transfer of the transfer starwheel centre of the upstream machine to the transfer of the transfer starwheel centre of the downstream machine + 5 clock pulses



Synonymous for the transfer to the filler

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Filler:

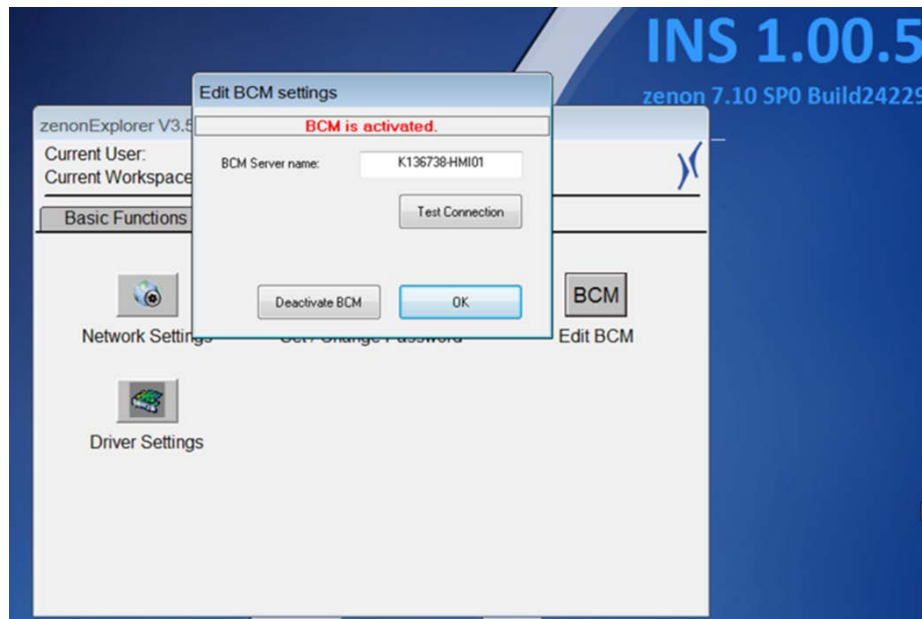
Check the DIP switches of the DP-DP coupler (filling technology only)

Set the parameter for "Machine empty": from the transfer of the upstream machine to the transfer onto the conveyor

Set the parameter "Transfer starwheel empty": see the "Block parameter list"

1.4 Visualisation system

Starting from the Zenon touch-screen Image INS 1.00.5, there is an aid available that enables automatic creation of the BCM project.ini file. For this, in the Zenon Explorer the name of the BCM server (usually the filler) must be entered in the Settings tab (e.g. K123456-HMI01).



1.5 Time synchronisation

The time server for the touch-screen computer in the ErgoBloc is not necessarily the filler!

According to R&D memo 2011/29, as from series version 04/2011 the time server is to be stored in the CMS list (<http://zmssrv1/>) (see the example screen, last line)

NTP time server: XXX.XXX.XXX.XXX

This address is to be set on all touch-screen computers as the time server

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◀◀ Suche ändern
◀◀ Erste Seite
◀ Zurückblättern
▶ Weiterblättern

Netzwerkdetails							
Nr	Kunde Linie	Hosts	Typ	Netzwerk (Broadcast) Bereich Subnetzmaske	Netzwerktyp VPN Verbindung Netzwerkstatus	SAP-Nr Anlagen-Nr Netzwerkcommission	
15468	PT. AQUA GOLDEN MISSISSIPPI, JAWA BARAT (Nr. 4723)	58		10.21.225.128 / 25 (10.21.225.255) 10.21.225.129 - 10.21.225.254 255.255.255.128	Data line network EzVPN freigegeben (released)	7024278 44157231 C364114 K597X12	Details
Vers	Kommentare und Bemerkungen						
v1.3	Ergobloc 31.500 bph						

VLAN Information: ID: 103 Name: VLAN_103
 NCP Tunnelendpunkt WLAN-SSID: WPA-Schlüssel:
 Beschreibung:
 NTP-Time-Server: 10.21.225.129

◀◀ Suche ändern
◀◀ Erste Seite
◀ Zurückblättern
▶ Weiterblättern

A machine's own IP address is never to be set as the time server!

Further information on this is to be found in the service instructions "Time synchronisation NTP tiime server" TD10003821

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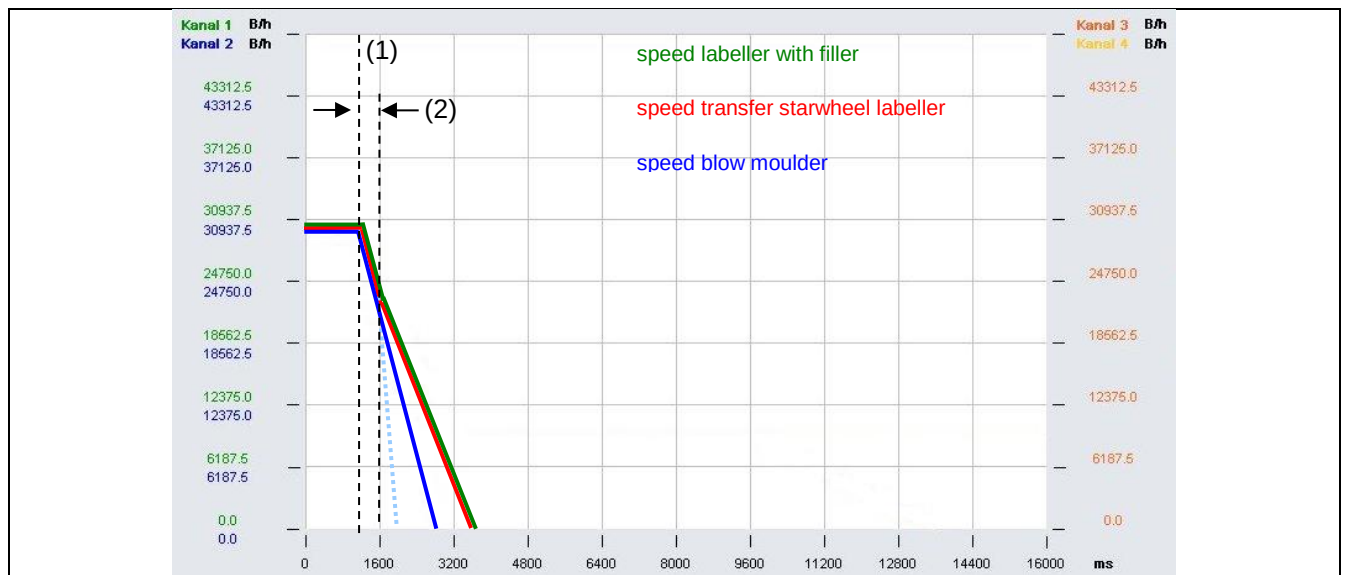
2 Block-spanning safety technology test, blocked machines

2.1 Block-spanning safety technology test

Test of the safety technology according to the valid AS-I Commissioning manual of the respective PL with visual inspection

Note on the points to be tested:

- EMERGENCY STOP between the machines
 - Operate all three machines at maximum speed
 - Confirm ESTOP at an arbitrary machine
 - Servo joint control is initiated between the blow moulder and the labeller
 - The labeller and the filler do not decouple from each other!
 - After the decoupling operation, the labeller (filler coupled to the labeller) and the blow moulder decelerate independently of each other
 - No servo drive malfunction must occur
 - Safe disconnection of the transfer starwheel through "Guards opened" on the upstream machine
- Safe disconnection of the transfer starwheel by partitioning
- Activation of "Safe speed SLS" through "Guards opened" on the upstream machine
 - Disconnection of the height adjustment drive Vortec base cooling system by means of "SLS active K4" from the labeller



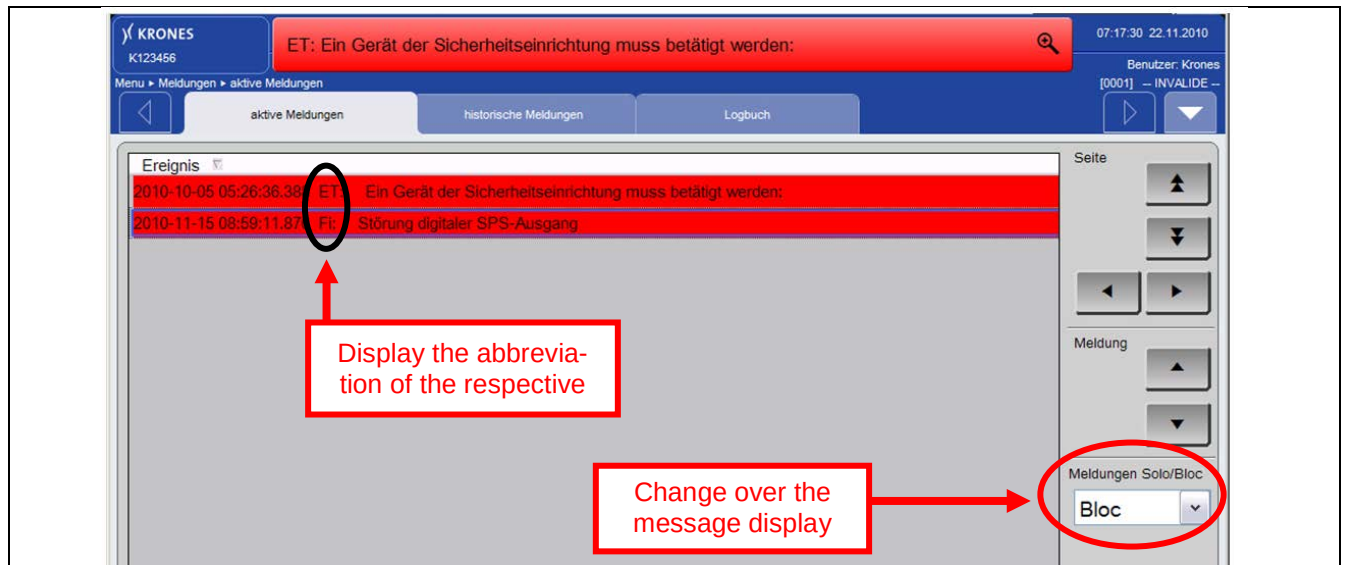
2.2 Block-spanning messages

Check on the touch-screens of the respective machines whether the messages are also displayed over the entire block (block-spanning)

To do so, change over the message display mode from solo to block and if necessary simulate the messages of a single machine

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Check the following features for plausibility:

- Date/time
- Machine abbreviation (BM; ET; FI)
- Message type
- Display of the message texts for language change-over

2.3 Signal exchange test

Inspection criteria:

- Check the signals related to the order no. based on the signal exchange list of the LDG drive (grey out unused signals)
- The signal designations must correspond exactly with those in the "Signal exchange list"
- Function-related tests on the signals must be checked immediately if possible (e.g. switch the mode of operation selector switch, do not force the signal)
- The status ("0" or "1") of the individual signals must be compared between the machines
Check the signals related to the order no. based on the signal exchange list of the LDG drive (grey out unused signals)

2.3.1 Signals from the blow moulder to the filler

Signals from the blow moulder to the filler tested successfully

2.3.2 Signals from the filler to the blow moulder

Signals from the filler to the blow moulder tested successfully

2.3.3 Signals from the filler to the labeller

Signals from the filler to the labeller tested successfully

2.3.4 Signals from the labeller to the filler

Signals from the labeller to the filler tested successfully

2.3.5 Signals from the labeller to the blow moulder

Signals from the labeller to the blow moulder tested successfully

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2.3.6 Signals from the blow moulder to the labeller

Signals from the blow moulder to the labeller tested successfully

2.4 JOG the block with all machines

2.4.1 JOG (in succession) the blow moulder, labeller and filler with the guards closed

- Synchronisation filler to the blow moulder
- For the ErgoBloc synchronisation filler to the labeller and labeller to the blow moulder
- All three machines rotate in sync with each other
- No JOG time limit on any JOG buttons

2.4.2 JOG with opened guards - blow moulder guards open

- JOG blow moulder (set-up mode: JOG button assigned according to the guards area)
- All machines rotate up to time limit
- No malfunctions must occur during the deceleration of the drives! (Note: ramp times)
- JOG downstream machines: no movement
-

2.4.3 JOG with opened guards - labeller guards open

- JOGGING at the labeller
- All machines rotate up to time limit
- No malfunctions must occur during the deceleration of the servo drives!
- JOGGING at the blow moulder and filler: no movement

2.4.4 JOG with opened guards - filler guards open

- JOGGING at the filler
- All machines rotate up to time limit
- No malfunctions must occur during the deceleration of the servo drives!
- JOGGING at the blow moulder and labeller: no movement

2.4.5 Partial block operation JOG with opened guards - guards (blow moulder open, partial block operation blow moulder - labeller)

- JOGGING at the blow moulder
- Blow moulder and downstream machine rotate up to time limit, if the guards of the downstream machine are closed
- No servo drive malfunction must occur
- Repeat the JOGGING test with opened guards at the labeller (the guards of the upstream machine are closed)

2.4.6 Partial block operation JOG with opened guards - labeller guards open (partial block operation labeller - filler)

- JOGGING at the labeller
- Labeller and filler rotate up to time limit, if the guards of the downstream machine are closed
- No malfunctions must occur during the deceleration of the servo drives
- Repeat the JOGGING test with opened guards at the filler (if the guards of the upstream machine are closed)

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2.5 Test the servo joint control block with all machines

How to push out all the parameters to set the fine clock pulses until the transfer is clear, as well as the clock pulses to empty the transfer starwheels is to be found in the document in the Intranet with the link indicated below

Departments...Bottling and packaging equipmentBottling technologyBlock and conveyor technologyBlock technology Engineering and Order Processing Block

2.5.1 Start

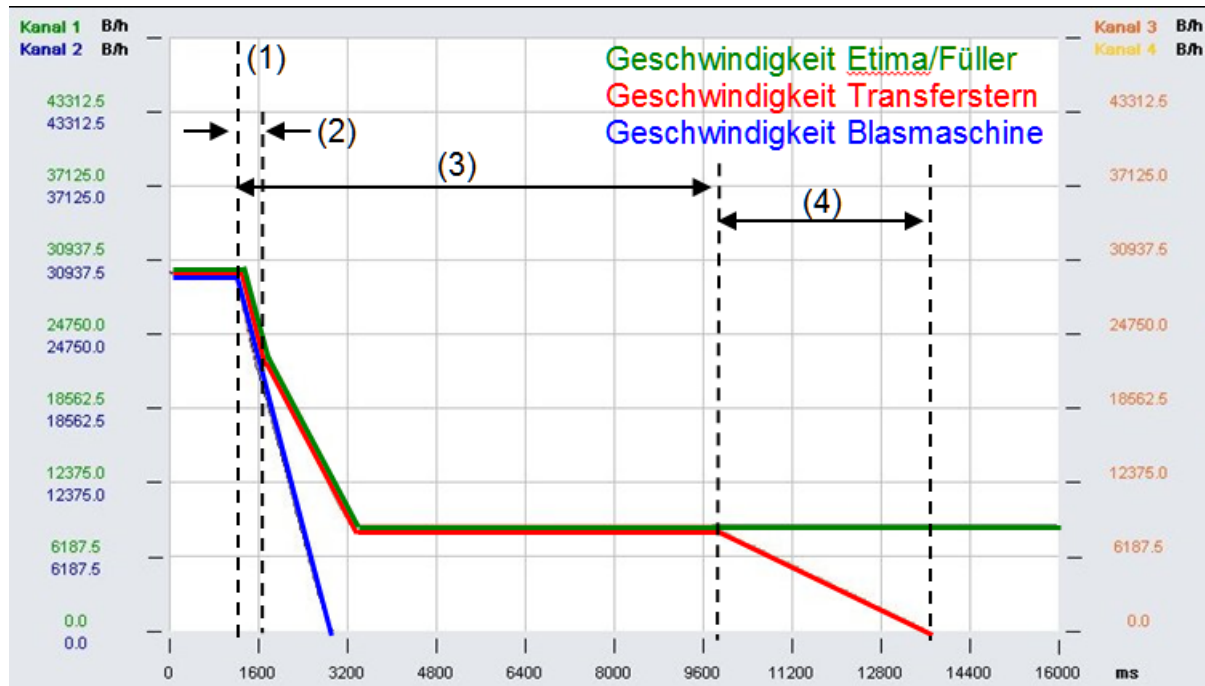
- Initial situation: all three machines are stopped
- Pressing a machine ON button on the blow moulder, labeller or filler
- The downstream machine synchronises itself with the lead machine
- For the ErgoBloc: The last machine synchronises itself to the centre machine and then the centre machine to the blow moulder

2.5.2 Blow moulder malfunction

- Initial situation: All three machines are operating at minimum production speed (see machine card)
- Simulation of a malfunction with an ESTOP at the blow moulder (1)
- Decoupling of the blow moulder from the downstream machine starts:
 - The blow moulder rejection unit pushes out containers and "the blow moulder executes the ESTOP ramp"
 - Waiting until the transfer between the blow moulder discharge starwheel and the labeller transfer starwheel is clear (fine clock pulses) (2)
 - The downstream machine changes over to the internal set-point value (request for synchronous operation = 0).
 - Waiting until the transfer starwheel is empty (after the starwheel has been pushed clear) (3)
 - The transfer starwheel turns OFF (4)
 - The blow moulder and the transfer starwheel are stopped
 - Open (unlock) the guards on the blow moulder comes
- Open the guards on the blow moulder
- Set the blow moulder mode of operation switch to "set-up" and JOG the blow moulder with the guards open
- The downstream machines continue operation
 - with production speed until the downstream machines are empty, then with minimum speed
- Eliminate the malfunction at the blow moulder
- Close the guards on the blow moulder
- RESET the malfunction and set the blow moulder mode of operation switch to "production"
- The "Machine ON" illuminating button is OFF on the blow moulder, the downstream machines blink
- Pressing a machine ON button (BM, LB, FU)
- The blow moulder starts
- The downstream machine synchronises itself with the blow moulder
 - If the downstream machines are empty, synchronisation with the blow moulder starts immediately
 - If the downstream machines are not empty, the labeller and filler are synchronised if the speed of the blow moulder is greater than that of the downstream machine
- The blow moulder rejection unit goes back if the labeller is in sync with the blow moulder

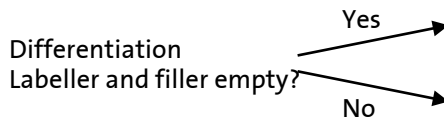
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Malfunction of the blow moulder decoupling operation

The labeller synchronises itself immediately after pressing of a machine ON button at the blow moulder



The labeller synchronises itself with the blow moulder if the speed of the blow moulder is greater than that of the labeller

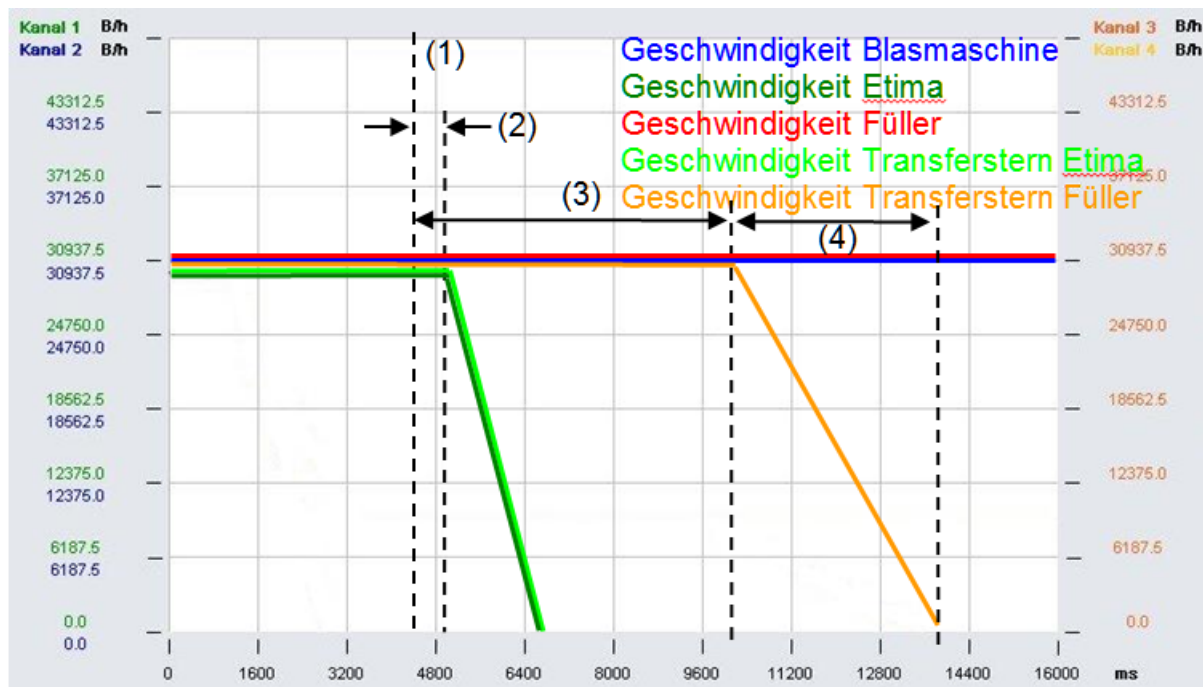
2.5.3 Labeller malfunction

- Initial situation: All three machines are operating at minimum production speed (see machine card)
- Simulation of a malfunction with ESTOP from the labeller starts: (1)
- Decoupling of the blow moulder from the labeller and of the filler from the labeller
 - The rejection unit transfer to the transfer starwheel of the blow moulder pushes out containers
 - Waiting until the transfer between the blow moulder discharge starwheel and the labeller transfer starwheel is clear (2)
 - The labeller and the labeller transfer starwheel execute the "ESTOP ramp"
 - The blow moulder empties itself and then continues to operate at minimum speed
- The decoupling of the labeller and the filler starts simultaneously with the decoupling of the blow moulder and the labeller
 - The rejection unit transfer to the filler pushes out containers
 - Waiting until the transfer between the labeller discharge starwheel and the filler transfer starwheel is clear (2)
- The filler changes over to the internal set-point value
 - If the filler is empty it operates at minimum speed
 - If the filler is not empty it operates at production speed until it is empty
 - Waiting until the transfer starwheel of the filler is empty
 - Filler transfer starwheel turns OFF
- Unlock the guards on the labeller
- Open the guards on the labeller
- Eliminate the malfunction on the labeller

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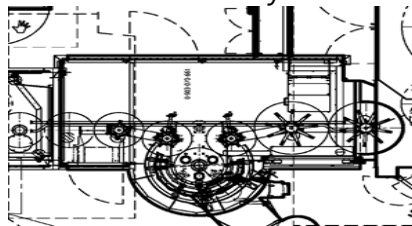
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- Set the labeller mode of operation switch to set-up and JOG with the guards open
- Blow moulder and labeller continue to operate
- Close the labeller guards and RESET
- RESET the malfunction and set the labeller mode of operation switch to production
 - If the filler is empty, it synchronises itself with the labeller and the "Machine ON" illuminating button at the filler blinks. The filler turns OFF after approx. 10 s
 - If the filler is not empty it continues operating at production speed until it is empty. The "Machine ON" illuminating button at the filler blinks
- The "Machine ON" illuminating buttons at the blow moulder and at the filler blink
- Pressing a machine ON button (BM, LB, FU)
- The labeller synchronises itself with the blow moulder if the filler is in sync with the labeller
- The blow moulder stop opens already during the synchronisation
- The blow moulder rejection unit goes back if the filler and labeller are in sync with the blow moulder



NOTICE!

The rear guard door of the labeller must always remain locked!
Reason: Personal safety! Partition is opened and the blow moulder is still operating!

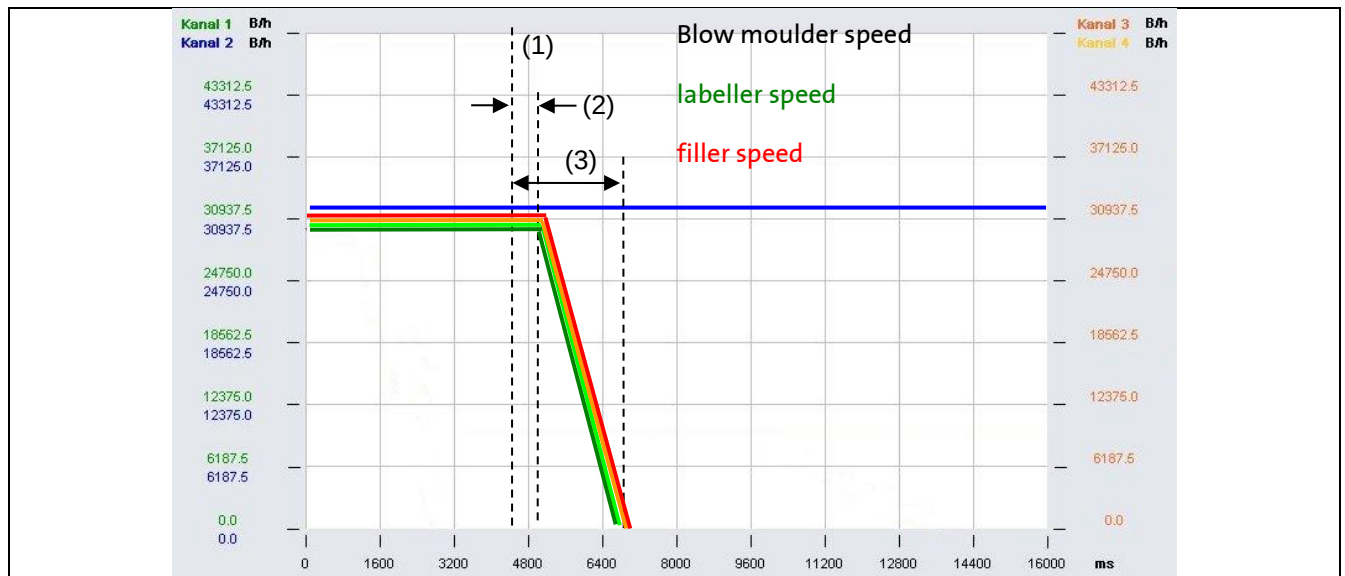


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2.5.4 Malfunction at the filler

- Initial situation: All three machines are operating at minimum production speed (see machine card)
- Simulation of a malfunction with ESTOP at the filler (1)
- Decoupling of the blow moulder from the labeller starts (no decoupling between the labeller and the filler!):
 - The blow moulder rejection unit pushes out containers; the blow moulder does not stop
 - Waiting until the transfer between the blow moulder discharge starwheel and the labeller transfer starwheel is clear (2)
 - The labeller changes over to the internal set-point value (synchronous operation = 0)
 - The filler, labeller and labeller transfer starwheel execute an "ESTOP ramp" (no decoupling between the labeller and filler!) (3)
 - The blow moulder empties itself and then continues to operate at minimum speed; the filler is stopped
 - The labeller transfer starwheel, the labeller, the filler transfer starwheel and the filler turn OFF
 - Open (unlock) the guards on the filler comes



Filler decoupling operation malfunction

- Open the guards on the filler
- Eliminate the filler malfunction
- Set the filler mode of operation switch to set-up and JOG the filler with the guards open (the labeller rotates together with the filler)
- Close the filler guards
- RESET the malfunction and set the filler mode of operation switch to production
- The "Machine ON" illuminating button on the blow moulder blinks; on the filler and the labeller it is OFF
- Pressing a machine ON button (BM, LB, FU)
- The filler and labeller synchronise themselves with the blow moulder
- The preform stop of the blow moulder opens already during synchronisation
- The blow moulder rejection unit goes back if the labeller and filler are in sync with the blow moulder

2.6 Test the operating statuses of the machine and of the signal beacon post (not necessary for EEIB)

2.6.1 Operating status

Company standard 1-099-00-610-0

LDS Operating status starting from Zenon 7.10 without state machines

2.6.2 Signal beacon post

Company standard 1-099-00-500-0

Conventional operating and signalling devices

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2.7 Block lead machine movement (not necessary for EEIB)

Block movement of the lead machine and downstream machine for

- partition malfunction
- Guards on the rear side of the labeller opened

2.8 Guards on the rear side of the labeller (not necessary for EEIB)

Unlocking of the guard doors not possible if after SJC the labeller is stopped and either the blow moulder or the filler is in operation

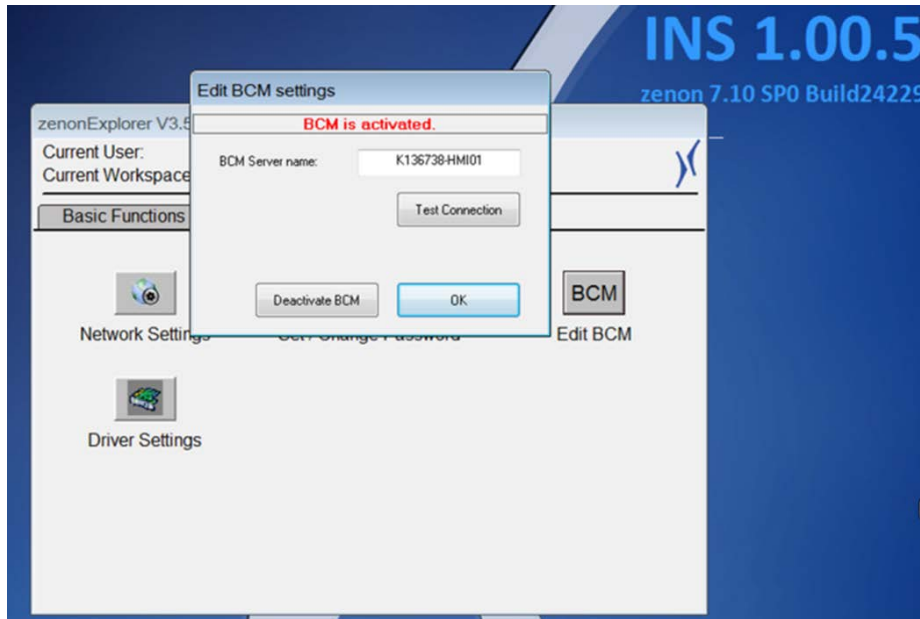
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3 Functional test with blocked machines in-house

3.1 Setting of the BCM server (visualisation BlocControl-Modul)

Starting from the Zenon touch-screen Image INS 1.00.5, there is an aid available that enables automatic creation of the BCM project.ini file. For this, in the Zenon Explorer the name of the BCM server (usually the filler) must be entered in the Settings tab (e.g. K123456-HMI01).



3.2 Functional test of the block-spanning safety technology

Necessary following a new machine installation:

Test of the safety technology according to the valid AS-I Commissioning manual of the respective PL with visual inspection

3.3 Functional test of the block-spanning shift register

Parameterisation

An S7 functional description is to be found under document no. TD10008404

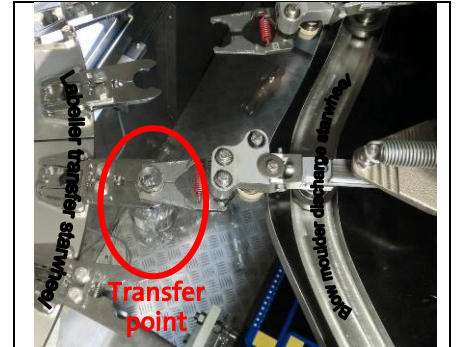
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3.3.1 Defining the blow moulder/labeller shift register transfer point

Fit the preform in the blow moulder (preceding the bottle present) and move/JOG it up to the transfer point to the labeller (Before rejection, lift the preform out of the clamp, continue to JOG briefly and then fit the preform again)

Read off the current position from the touch-screen and enter in Conti-form – Basic parameters – Discharge: "Transfer bottle to next machine in block".



Transfer bottle to the next machine in the block

Number of fine clock pulses until the transfer area is empty

3.3.2 Teach the labeller shift register

Prerequisites:

- Shift register transfer point blow moulder/labeller set
- All machines must be empty
- Block active

Proceed as follows:

- Turn ON the block
- On the labeller, call the page for parameterising the shift register:

Navigation: KRONES --> Parameters --> Machine parameters --> Machine basic settings --> Teach shift register transfer position

- Activate "Learn mode"
- Run a bottle through the blow moulder (only one bottle!)
- Manually take over the values from the "Current" column into the "Entry" column

	Aktuell	Eingabe	Zuletzt ermittelte Wert
Behälter vorhanden von Vorgängermaschine erhalten	0		
Einschreibeposition Feintakt ermittelt	2 FT		0 FT
Einschreibeposition Feintakt ermittelt	6 FT	0 FT	0 FT
Einschreibeposition ins Schieberegister	31	0	0
Schieberegisterposition Behälter vorhanden (Offset)	100	50	0
Position Behälterübergabe am Auslauf	124 GT	0 GT	

3.3.3 Teach the filler shift register

Prerequisites:

- Shift register transfer point labeller/filler (to the labeller) set
- All machines must be empty

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- Block active

Proceed as follows:

- Turn ON the block
- On the filler, call the page for parameterising the shift register:

Navigation: KRONES --> Parameters --> Machine parameters --> Machine basic settings --> Teach shift register transfer position

- Activate "Learn mode"
- Run a bottle through the labeller (e.g. fit it before bottle present; only one bottle!)
- Manually take over the values from the "Current" column into the "Entry" column

The screenshot shows the 'Schieberegister Übergabeposition lernen' (Teach shift register transfer position) screen. At the top, there's a status bar with a red warning: 'Durchflussmessgerät ist gestört: FIT 700'. Below this is a navigation menu with options like 'Füller Ziel', 'Schraubverschleißer (2)', 'Schraubverschleißer Sortierwerk', and 'Schieberegister Übergabeposition lernen'. The main area has a 'Modus Lernen' section with a power icon and a 'I' icon. Below this is a table with four columns: 'Aktuell' (Current), 'Eingabe' (Entry), and 'Zuletzt emittierter Wert' (Last emitted value). The table contains five rows of parameters related to the shift register and bottle transfer.

	Aktuell	Eingabe	Zuletzt emittierter Wert
Behälter vorhanden von Vorgängermaschine erhalten ⚠	0		
Einleseposition Feintakt emittiert ⚠	0 FT		0 FT
Einschreibeposition Feintakt emittiert ⚠	0 FT	0 FT	0 FT
Einschreibeposition ins Schieberegister ⚠	0	0	0
Schieberegisterposition Behälter vorhanden (Offset) ⚠	0	150	0

At the bottom, there's a status bar with 'online: [0001] Maschineneinstellung', 'Speichern' (Save) button, and 'Verlassen' (Leave) button.



A detailed description on setting the block-spanning shift register can be called by pressing the button.

3.3.4 Teach splicing Controll labelling station

Prerequisites:

- Shift register transfer point blow moulder/labeller (to the labeller) set
- All machines must be empty
- Block active

The pitches from the transfer point of the bottle to the labeller up to the transfer of the label to the bottle must be added to the value of the no. of customer objects for emptying the blow moulder (see item 3.4 Testing the empty situation) This is the parameter interval container stop labelling station

Proceed as follows:

- On the labeller, call the page "CR splicing" and enter the following settings:

Navigation: KRONES -->Parameters -->Machine parameters -->Contiroll -->Machine parameters -->CR splicing

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Verklebung innerhalb einer Lücke

Dauer Lücke 3 s

Abstand Behältersperre <-> Aggregat 245

Gemessener Abstand Behältersperre <-> Aggregat 1482

- Duration of the gap => 3 s
- Calculate the container stop distance <-> labelling station => shift register value from the container stop of the blow moulder to the Controll labelling station 1 of the labeller (see item 8, no. of customer objects in the blow moulder)
- Prepare splicing
- Turn ON the block
- Release the container stop
- After approx. 20-30 preforms, press the button "Manual splicing" at the labelling station



- The stop closes (container gap) and opens again after a time of 3 s
- Close the container stop via the "manual" button at the touch-screen
- The splicing should be positioned in the centre of the gap

3.3.5 Test

Run a bottle through the entire block and observe the display in the shift registers. Only a "10" may appear in each respective shift register.

If the functional test is not successful, the parameterising process must be performed again.

3.3.6 Simulation of a malfunction

Simulation of a labeller malfunction:

- Run a cluster of bottles through the blow moulder
- After transfer to the labeller, effect a stop (use the labeller function "Test container" - see TD12000765)
- Remove two bottles preceding the labeller bottle present
- Turn ON the block again
- Warning "Container transfer monitoring" and check the signal "Enable container stop release"
- RESET the malfunction and repeat the process
- Remove five bottles preceding the labeller bottle present
- Check the malfunction "Container transfer monitoring consecutive fault"

Simulation of a filler malfunction:

- Run a cluster of bottles through the blow moulder
- After the transfer to the filler, effect a stop
- Remove two bottles preceding the filler bottle present
- Turn ON the block again
- Warning "Container transfer monitoring" and check the signal "Enable container stop release"
- RESET the malfunction and repeat the process
- Remove five bottles preceding the filler bottle present
- Malfunction "Container transfer monitoring consecutive fault", partition between the blow moulder and the labeller

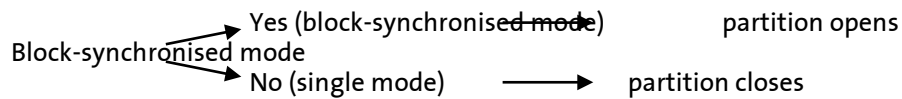
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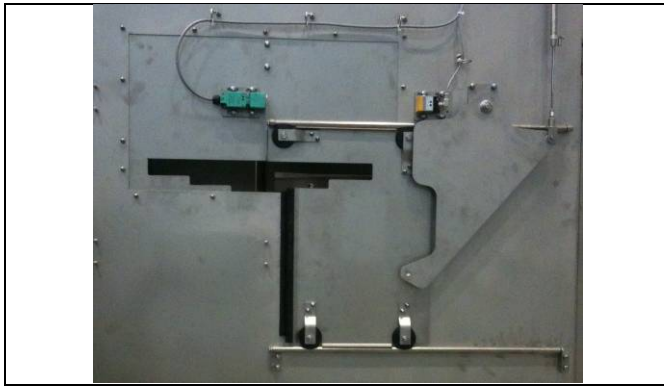
3.4 Functional test of the partition between the machines

Partition between the blow moulder and the labeller

3.4.1 Functional test



- Optical monitoring of the mechanical leverage of the installed cylinder
- The partition must be able to rise and descend without any great resistance
- Movement of the partition can be stopped by hand at any time
- In assembled condition, there must be no gap when the partition is closed
-



3.4.2 Simulation of a malfunction

Proceed as follows: To do so, operate the machine at **maximum** speed and push the partition by hand to the centre position so that no proximity switch is actuated

- Check the message "Faulty position partition"
- There must be no malfunction at the transfer starwheel!

3.4.3 Block machine movement of the upstream machine

As long as the partition is in the centre position, neither the labeller nor the blow moulder can be JOGGED or turned ON

Partition between the labeller and the filler

3.4.4 Functional test

The test is performed in a similar manner to 3.3.1

3.4.5 Simulation of a malfunction

The test is performed in a similar manner to 3.3.2

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3.4.6 Block machine movement of the upstream machine

The test is performed in a similar manner to 3.3.3

3.5 Testing the empty situation

Set the parameters for "Machine empty": Parameter for preforms at the infeed P.E. sensor + parameter clock pulse for the transfer of shift register information + 5 clock pulses

The "Machine empty" parameter is set on each individual machine such that the individual machines are empty
The block does not switch to minimum speed until all the block-synchronised machines are empty

3.6 Functional test of a power failure in block-synchronised mode

For this test it is assumed that a power failure test was already conducted on the individual machines and that it functioned. Check the following points first without bottles!

Functional test of a power failure in block-synchronised mode

Blow moulder power failure

- Operate the block at maximum production speed
- Turn OFF the master switch of the blow moulder
- The blow moulder rejection unit must intercede immediately and reject the bottles
- As far as possible, visual inspection as to whether the transfer areas are free
- Immediately after the power failure is detected, the blow moulder shuts down with the ESTOP ramp
- The labeller and the filler detect the communication malfunction or power failure via Powerlink and also shut down shortly afterwards with the ESTOP ramp (they should however remain in sync)

Labeller power failure

- Operate the block at maximum production speed
- Turn OFF the master switch of the labeller
- The rejection units of the blow moulder and filler must intercede immediately and reject the bottles
- As far as possible, visual inspection as to whether the transfer areas are free
- Immediately after the power failure is detected, the labeller shuts down with the ESTOP ramp
- The blow moulder and the filler detect the communication malfunction or power failure via Powerlink and also shut down shortly afterwards with the ESTOP ramp

Filler power failure

- Operate the block at maximum production speed
- Turn OFF the master switch of the filler
- The rejection units of the blow moulder and labeller must intercede immediately and reject the bottles
- As far as possible, visual inspection as to whether the transfer areas are free
- Immediately after the power failure is detected, the filler shuts down with the ESTOP ramp

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- The blow moulder and the labeller detect the communication malfunction or power failure via Powerlink and also shut down shortly afterwards with the ESTOP ramp

Power failure of the entire block-synchronised machine

- Operate the block at maximum production speed
- Turn OFF the master switches of the three machines as simultaneously as possible
- The rejection units in front of the transfer starwheels must intercede immediately and reject the bottles
- All three machines shut down independently in sync (no coupling)

Power failure of the entire block machine with bottles

- Operate the block at maximum production speed
- Turn OFF the master switches of the three machines as simultaneously as possible
- The rejection units in front of the transfer starwheels must intercede immediately and reject the bottles
- All the machines detect the communication malfunction or power failure via Powerlink and shut down with the ESTOP ramp (No coupling)
- Check the transfer areas for damages to the grippers and bottles

3.7 Control horn in case of lack of POM (all the machines in the block)

When the processing and operating materials (POMs) are coming to an end, the blue lamp blinks as a warning signal. On the ErgoBlocL, the filler also triggers a horn.

If all three machines are in block-synchronised mode and one machine reports POM shortage, the horn must sound as a warning signal.

The signal exchange is effected via Profibus (see "Signal exchange test").

Functional test:	
Filler:	Lack of caps
Labeller:	Lack of labels
Blow moulder:	Lack of preforms

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4 With blocked machines on site

4.1 Functional test of block-spanning safety technology

Necessary after a new machine installation and if not already tested with item 3.2:

Test of the safety technology according to the valid AS-I Commissioning manual of the respective PL with visual inspection

4.2 Torque recording during operation

The recording of the torques must take place already during the commissioning of the individual machines

Checking of the maximum torques during operation (above all at the transfer starwheels)